

BINF6999 Intermediate Reflection: When Zero Is an Answer

Project: A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing
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What

My project asks why tendon healing goes one of two ways: regeneration, where the tissue recovers its structure and function, or fibrosis, where a permanent collagen scar forms instead. The outcome depends largely on which of two progenitor cell populations wins out at the injury site. Tendon Stem/Progenitor Cells (TSPCs) drive regeneration; Tendon Fibro/Adipogenic Progenitors (T-FAPs) drive fibrosis. Both live in the same tendon sheath niche, and the question I am trying to answer is what transcription factors (TFs) control which fate a progenitor cell takes.

In mid-June, I ran pySCENIC (Van de Sande et al., 2020), a computational pipeline that infers TF regulatory networks from single-cell RNA sequencing data, on two independent datasets. A regulon, the unit pySCENIC outputs, is a transcription factor paired with the set of genes it controls, scored per cell. I ran this on the Harvey 2019 dataset (4,069 cells from uninjured adult patellar tendon) and the Cherief 2023 dataset (22,615 cells from injured Achilles tendon under two conditions: innervated, where sensory nerves are intact and healing proceeds normally, and denervated, where sensory nerves are blocked and healing shifts toward fibrosis). The pipeline follows three steps: GRNBoost2 builds co-expression networks between TFs and target genes, cisTarget prunes those networks to keep only TF-gene links supported by known DNA binding motifs, and AUCell scores how active each regulon is in each individual cell. Both runs completed without errors, producing 471 regulons from Harvey and 417 from Cherief.

The plan from there was to apply a cross-dataset direction filter: keep only regulons where statistical testing showed enrichment in the same cell type (either TSPC or T-FAP) in both datasets independently. The idea was that whatever survived would reflect genuine cell-type identity programs rather than noise or dataset-specific artifacts. This filter was central to the mechanistic arm of the project.

The filter returned **15 consistent T-FAP regulons** (a coherent AP-1/KLF/NF- κ B inflammatory and fibroblast-activation program anchored by Junb, Fos, Fosb, Klf6, Klf9, Nfkb1, Egr1, Atf3, and Cebpd) and **zero consistent TSPC regulons**. Harvey 2019 did show TSPC-enriched TF activity: E2f1 (cell cycle), Fli1 (ETS family), and Arnt2 (bHLH progenitor programs) were all significantly enriched in TSPCs there. But in Cherief 2023, those same regulons were reversed or absent in TSPCs and so none of them made it through.

No TSPC regulons passed both filters.

So What

My first instinct was that something had gone wrong: a misconfigured threshold, a batch effect, maybe just not enough TSPC cells (the sub-cluster had 451). I went back through the heatmaps, checked per-cell regulon score distributions, and re-examined the sub-cluster annotations. Nothing pointed to a technical failure. The zero was real.

That forced a question I hadn't thought to ask before running the analysis: does cross-dataset TF regulon consensus even make sense when the two datasets come from completely different tissue states?

Harvey 2019 is uninjured adult patellar tendon. TSPCs there are in a homeostatic resting state, quiescent but cycling; E2f1 and other cell-cycle programs are active. Cherief 2023 is day-14 post-injury Achilles tendon saturated with inflammatory signals. TSPCs in that context are dealing with an entirely different environment, and their transcriptional state reflects that. The T-FAP AP-1/KLF program, by contrast, shows up in both datasets because activated fibroblast biology gets switched on by injury regardless of tissue site. It is a generic injury response that T-FAPs happen to lock into persistently. TSPC identity is more context-sensitive.

What bothered me most was how clearly the result exposed a flaw in my original assumptions. The proposal had framed pySCENIC as a tool that would return a symmetric output: a TSPC program and a T-FAP program, which would then serve as parallel biological features for a classifier predicting healing outcome. That framing treated TF regulon activity as though it were like a protein marker, stable and cell-type-intrinsic across any dataset containing that cell type. But TF programs reflect functional state, not identity. The same cell in two different environments can be running completely different transcriptional programs, which is actually the whole point of studying how injury shifts progenitor fate in the first place. Seeing it play out directly in the data was a different experience from understanding it conceptually.

There is also a broader issue that this surfaced. Cross-dataset TF inference is increasingly used as a validation strategy in single-cell biology, where the reasoning is that a TF program appearing in two independent datasets must be real biology. That reasoning holds when the datasets represent comparable tissue states. When they do not, asymmetric results are biologically expected, and treating them as methodological failures rather than findings causes you to miss what the comparison is actually showing.

Now What

Before this, I would have treated an asymmetric cross-dataset result as a problem to fix, either by adjusting parameters or questioning data quality. Now I think the first question has to be whether the two datasets are even asking the same biological question. Here, they weren't. Harvey 2019 captures what TSPCs look like at homeostasis; Cherief 2023 captures what they look like in an inflammatory injury environment. The filter was working correctly. It just couldn't recover what wasn't consistently there to recover.

Recognizing that changed what I could claim from the pySCENIC output, and it also changed how I approached feature selection for the classifier. Rather than a symmetric TSPC/T-FAP TF-target feature space, I restructured the candidate gene set so the 15 T-FAP regulon targets serve as positive fibrotic features, while TSPC identity is captured through marker genes with stable expression across tissue states (Tppp3, Prg4, Igfbp6, Sema3c, Cd34). I also combined the pySCENIC results with output from CellRank, a separate tool that models which genes are most associated with a cell moving toward a particular fate. The overlap between pySCENIC T-FAP regulon targets and CellRank T-FAP fate-driver genes produced 119 candidate genes (dominated by Junb, Klf9, and Klf6 targets) that are now the primary input for machine learning feature selection using LASSO, Boruta, and XGBoost.

The classifier validation also needed rethinking. A classifier trained on homeostatic Harvey 2019 labels cannot be validated against Cherief 2023 as a straight replication test since the tissue states differ by design. The more meaningful test is directional: denervation is known to shift healing toward fibrosis, so a well-calibrated classifier should shift toward the fibrotic end of the score distribution in denervated samples. That is a hypothesis about biological behavior, not a check for numerical agreement, and it is a more honest framing for what the classifier is trying to do.

The pySCENIC result took a while to stop reading as a failure. Once I did, it turned out to be the most clarifying result of the first half of the project, because it forced me to be explicit about something the analysis had been quietly assuming all along.